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1. Measurement, Validity, and Reliability in Political Inquiry

1.1 The Strategic Imperative of Measurement

- **The Harsh Truth**
    - Political arguments only as robust as supporting data.
    - Measurement = Strategic bridge between "Theoretical Plane" (ideas) and "Empirical Plane" (reality).
    - Without measurement, political science reduces to philosophy.

1.2 From Theory to Observation: The "So What?"

- **Defining the Battlefield:** Moves theory into the realm of falsifiability; unmeasurable theories are unscientific.

- **Rejection of Subjectivity:** Shifts from personal "conceptions" to "inter-subjective" agreement (e.g., agreeing on visible indicators of Democracy).

- **Precision and Replicability:** Standardization allows independent verification across time and space.

- *Goal:* Mastery of mental discipline before application of statistical tools.

1.3 From "Fuzzy" Ideas to Concrete Constructs

- **The "Original Sin" of Research**

    - Failure to define terms (e.g., Democracy, Power) leading to ambiguity.

    - **Conceptualization:** Defining abstract/fuzzy ideas in precise terms.

1.4 The "Scientific Lie": Making Abstractions Concrete

- **Reification**

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  * Treating mental abstractions (constructs) as if they were real, tangible objects.

    * A "scientific lie" necessary for measurement.
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- *The Risk:* Flawed conceptualization leads to studying illusions.

1.5 The Anatomy of a Construct

- **Unidimensional:** Single underlying scale (e.g., Weight).

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  *Risk:* Oversimplification (e.g., treating Self-Esteem as single-faceted).
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- **Multidimensional:** Multiple underlying dimensions (e.g., Academic Aptitude = Math + Verbal).

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    *Risk:* Measurement error if dimensions are ignored.
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1.6 Operationalization and Measurement Scales

- **Operationalization:** The blueprint for data collection; dictates statistical handling.

1.7 The Four Levels of Measurement

- **Nominal (Categorical):** Mutually exclusive labels; no order. (e.g., Regime Type). *Stats:* Mode, Chi-square.
- **Ordinal:** Rank-ordered; unknown distance between ranks. (e.g., Political Activism). *Stats:* Median, Percentiles.
- **Interval:** Rank-ordered + equidistant; arbitrary zero. (e.g., IQ Scores). *Stats:* Correlation, Regression (often applied to Likert).
- **Ratio:** Interval qualities + true zero. (e.g., Military Spending). *Stats:* All permissible.

1.8 Measurement Strategies for Constructs

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    - **Scale Strategy for surveys**
- **Likert Scales:** Intensity of agreement; technically ordinal but often treated as interval for regression.
- **Guttman Scales:** Cumulative intensity; hierarchy of engagement (High intensity agreement implies low intensity agreement).

1.9 Examples: Likert Scale Question

- "To what extent do you agree with the following statement: 'Democracy is the best form of government.'"
    - Strongly Disagree
    - Disagree
    - Neutral
    - Agree
    - Strongly Agree

1.10 Example: Guttman Scale Questions

- "Have you ever participated in the following political activities?"
    1. Voted in a general election

2. Voted in a primary election
3. Signed a petition
4. Attended a political rally
5. Volunteered for a campaign
6. Donated money to a political cause
7. Run for public office

1.11 4. The Yardsticks of Scientific Rigor

- **Standardization:** The defense against biased observation.

- **The Shooting Target Analogy**

  - **Reliability (Consistency):** Hitting the same spot repeatedly (even if off-center).
  - **Validity (Accuracy):** Hitting the bullseye; measuring what is claimed.

1.12 Evaluating Validity (The Pillars)

- **Internal Validity:** Establishing causality vs. spurious correlations.

- **External Validity:** Generalizability to broader populations

- **Construct Validity:** Ensuring the tool measures the actual concept

- **Statistical Conclusion Validity:** Appropriate use of mathematical tests for the data type

2. Application & Discussion: Kahan & Corbin (2016)

2.1 The Anatomy of the Study: Variables & Hypotheses

- **The Baseline Check**
  - What is the **Dependent Variable (DV)** in this study? (What is being explained?)
  - What is the **Independent Variable (IV)**? (What is doing the explaining?)
  - *Hint:* Look at the title. “Actively Open-Minded Thinking” (AOT) vs. “Polarization.”

2.2 The Theory vs. The Hypothesis

- **Theory:** What does the literature suggest *should* happen when people think more critically?

  - **Hypothesis:** If AOT measures “neutrality,” what should the slope look like?

  - **The Result:** What actually happened? (The “Perverse Effect”).

2.3 Critiquing the Construct: Measuring "Open-Mindedness"

- **Conceptualization Challenge**
  - How do you measure a thought process?
  - Kahan & Corbin use the "Actively Open-Minded Thinking" (AOT) scale.
  - *Class Discussion:* Is this scale measuring a *willingness* to change one's mind, or a *capacity* to argue better?
- **Validity Check**
  - **Face Validity:** Do the questions ostensibly look like they measure open-mindedness?
  - **Construct Validity:** If AOT correlates with *higher* polarization, is the measure invalid?

2.4 Thought Question

- *Provocation:* If a thermometer showed water freezing at 100°C, would you blame the water or the thermometer? (Apply this logic to the AOT scale).

2.5 Operationalizing "Polarization"

- **Defining the Term**
  - Is "Polarization" a behavior (shouting/protesting) or a mental state (belief divergence)?
- How do the authors measure polarization?

2.6 Measurement Strategy

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  - The authors measure polarization via "Climate Change Risk Perception."
    - *Critique:* Is "Risk Perception" a valid proxy for "Political Polarization"? Why or why not?
    - *Alternative Measures:* How else could we measure polarization?
      - Voting records? (Nominal)
      - Affective Thermometer ratings of the other party? (Interval/Ratio)

2.7 The Validity Paradox: Instrumental vs. Intrinsic

- **The "So What?" for Political Science**
  - The study shows that high AOT scores + Partisanship = *Maximum* Polarization.
- **Re-evaluating the Construct**

    - Does this mean the AOT scale has **Low Validity** (it failed to predict open-mindedness)?
    - OR… does it mean our **Theory** of open-mindedness was wrong?

2.8 Discussion

- What else could AOT be measuring besides open mindedness?

- **Cognitive Sophistication:** the ammunition used to defend the tribe, not a willingness to abandon it

- **Motivated Reasoning:** Ability to rationalize pre-existing beliefs more effectively

2.9 Internal Validity Threat

- **Spurious Correlation Risk**
    - Could a “Third Variable” be driving these results? (e.g., Education, Income, Media Consumption).

2.10 Transferring the Logic: Measuring “Democracy”

- **From Micro (Individuals) to Macro (Regimes)**
    - If measuring “Open-Mindedness” is this hard, how do we measure “Democracy”?

2.11 Conceptualization Exercise

- **Binary Strategy:** Is Democracy a switch? (Democracy vs. Autocracy).

- **Continuous Strategy:** Is Democracy a spectrum? (Polity IV score -10 to +10).

- **Operationalization Trade-offs**

    - If you include “Economic Equality” in your measure of Democracy, what happens?
    - *Risk:* You can no longer test if “Democracy causes Wealth” because you have built wealth *into* the definition of Democracy. (Tautology).

2.12 Wrap-Up and Final Takeaways

- **Integrity of Discipline:** Relies on transparency and rigor of measurement.

- **Statistical Complexity:** Cannot salvage poor conceptual design.

- **The Four Pillars of Scientific Method**

    - **Replicability:** Independent repetition must yield similar results.
    - **Precision:** Exact definitions allowing universal application.
    - **Falsifiability:** Theories must be capable of being disproven.

    - **Parsimony (Occam’s Razor):** The simplest explanation is superior.

2.13 Authorship, License, Credits

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